

Thermal Hydraulics for the Concentric Reactor Model

Modelica loop and fuel heat-structure note

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1 Purpose

This note documents the thermal-hydraulic model now used for the Reactor Kinetics Lab concentric reactor design. The model is not a licensing-grade subchannel calculation. It is a compact Modelica plant model intended to provide:

- primary-loop inlet and outlet coolant temperatures,
- primary mass flow and core pressure drop diagnostics,
- fuel and moderator effective temperatures for reactivity feedback,
- an effective D₂O moderator density for density feedback,
- a transient fuel heat capacity between fission power and coolant heat removal.

The key change from the earlier annular-core note is that the thermal model is now framed around the OpenMC concentric fuel element. The hydraulic circuit remains a single effective primary channel, but the fuel heat storage and heat-transfer area are derived from the resolved concentric fuel-ring geometry.

2 Concentric Reactor Inputs

The first-pass thermal model uses the OpenMC concentric fuel-element defaults as its core geometry source of truth:

Quantity	Value
Nominal thermal power P	20 MW
Fuel ring count	6
Fuel meat thickness	1.5 cm
Coolant gap between rings	4.5 cm
Fuel-element inner radius	20 cm
Fuel-element outer radius	60 cm
Active fuel height	350 cm
Fuel density	12.2 g/cm ³
Primary coolant/moderator	D ₂ O represented by the current water approximation

From these values, the aggregate fuel-meat quantities are

$$V_f \approx 0.7917 \text{ m}^3, \quad m_f \approx 9.66 \times 10^3 \text{ kg}, \quad A_{ht} = \frac{V_f}{t_f/2} \approx 105.6 \text{ m}^2,$$

where $t_f = 1.5 \text{ cm}$ is the fuel meat thickness. The implied average fuel-meat power density at 20 MW is approximately

$$q_f''' = \frac{P}{V_f} \approx 25.3 \text{ MW/m}^3.$$

The effective Modelica core pipe uses the OpenMC active height. Its diameter is intentionally retained as a calibrated single-channel hydraulic diameter, not the 1.2 m fuel-envelope outer diameter. The concentric OpenMC dimensions enter the fuel heat storage and heat-transfer area directly; pressure drop remains a coarse loop diagnostic rather than a resolved subchannel prediction.

3 Operating Envelope

The nominal primary-loop operating point is unchanged:

Quantity	Selected value
Primary inlet bulk temperature target	25°C
Nominal outlet bulk temperature target	45°C
Nominal bulk temperature rise	20 K
Nominal mass flow	237 kg/s
Core flow direction	top-to-bottom
Primary pressure regime	low pressure, single phase

At this stage the pump remains flow-controlled at the nominal mass flow. The inlet plenum is represented as a closed compliance volume connected to a fixed 4 bar pressure-reference boundary. This keeps the FMU hydraulic pressure level well posed during coupled transients and prevents unphysical absolute-pressure drift in the closed primary loop. The pressure drop is therefore a diagnostic for the current effective-pipe geometry, not a validated subchannel loss prediction. In particular, the current pipe hydraulic diameter is an effective calibration value used to keep the primary-loop pressure model well conditioned.

The inlet temperature of 25°C is set by the return temperature from the pool and the secondary heat exchanger. The nominal outlet of 45°C keeps the entire primary loop comfortably below saturation at any plausible low-gauge operating pressure and well within the single-phase D₂O regime. The 55°C upper bound is a conservative operating limit, not a safety margin; the actual saturation temperature at 3 bar gauge is approximately 133°C.

4 Reactor-scale flow and heat-removal estimates

4.1 Required primary mass flow

Using D₂O properties at the operating temperature midpoint ($\approx 35^\circ\text{C}$):

$$c_p \approx 4215 \text{ J}/(\text{kg K}), \quad \rho \approx 1100 \text{ kg}/\text{m}^3.$$

The energy balance gives:

$$\dot{m} = \frac{P}{c_p \Delta T}.$$

Case	ΔT	$\dot{m}$ [kg/s]	$\dot{V}$ [m ³ /h]
Nominal design point	20 K	237	776
Upper-bound design point	30 K	158	518

The nominal design flow of approximately 237 kg/s (776 m³/h) is the primary target for pump and piping sizing. The upper-bound flow of 158 kg/s sets the lower flow limit at which the outlet temperature constraint is still met.

4.2 Pump duty

The pump shaft power scales with the loop pressure drop:

$$W_{\text{pump}} \approx \frac{\Delta p \cdot \dot{V}}{\eta}, \quad \eta \approx 0.7.$$

For the nominal design point:

Loop Δp	W_{pump} (nominal $\dot{m}$)	W_{pump} (upper-bound $\dot{m}$)
1 bar	31 kW	21 kW
2 bar	62 kW	41 kW
3 bar	92 kW	62 kW
5 bar	154 kW	103 kW

A total loop pressure drop in the range of 2–3 bar is a plausible first estimate for a large-scale research-reactor primary circuit, giving a pump duty on the order of 60–90 kW at the nominal flow.

4.3 Pool thermal time constant

The surrounding pool (Section ??) has a volume of 2535 m³ (2.535×10^6 litres). Its thermal time constant at full reactor power with no active cooling is:

$$\tau = \frac{m_{\text{pool}} c_{p,\text{water}}}{P} = \frac{2535 \times 10^3 \times 4200}{20 \times 10^6} \approx 530 \text{ s} \approx 9 \text{ min.}$$

This confirms that the pool cannot serve as the primary heat sink for sustained operation: at full power it would heat up by 20 K in roughly 9 minutes. The primary forced-circulation loop is therefore the essential heat-removal path, and the pool is a structural heat reservoir for short transients only.

4.4 Flow direction

The primary coolant flows **top to bottom** through the effective core channel. The inlet is at the top of the active region; the outlet is at the bottom. This matches the natural downward-flow convention seen in several research-reactor designs and means the hottest bulk coolant leaves at the bottom before routing to the heat exchanger.

4.5 Stagnant reflector and pool

The outer D₂O reflector and the surrounding pool water together constitute a large stagnant inventory that provides:

- long-term thermal mass for decay-heat transient absorption,
- shielding of personnel and instrumentation,
- neutron moderation in the outer reflector zone.

The rectangular pool surrounding the reactor vessel has approximate dimensions of 15 m (height) $\times$ 13 m $\times$ 13 m (base), giving a total volume of approximately 2535 m³ (2.535×10^6 litres).

In the next Modelica model, this inventory is represented as a **fixed-temperature or slowly varying thermal boundary condition**. Active pool cooling, pool-loop flow, and pool temperature dynamics are not modelled in the first implementation pass.

5 Simplified primary loop

The primary D₂O loop follows a fixed topology inspired by forced-circulation research reactors such as FRM II, but simplified aggressively to the minimum useful set of components.

5.1 Topology

1. **Upper inlet plenum.** A closed compliance volume at the top of the core that distributes incoming primary coolant across the effective core channel entrance, provides hydraulic decoupling, and connects to a fixed 4 bar pressure-reference boundary for FMU pressure anchoring.
2. **Effective core channel.** A single axially discretised heated pipe (8 nodes) carrying the total primary flow top to bottom. Heat input follows the axial power profile from the neutronics model.
3. **Lower outlet plenum.** A vessel volume at the bottom of the core that collects the heated coolant before it exits to the primary piping.
4. **Primary pump.** A single centrifugal pump model with a representative head curve operating on the D₂O primary inventory. Nominal operating point: ≈ 237 kg/s at 2–3 bar head.
5. **Primary heat exchanger (primary side).** A counterflow shell-and-tube or plate-heat-exchanger model on the primary side that removes the full 20 MW to the secondary loop at nominal conditions.

5.2 Selected nominal operating parameters

Parameter	Value
Primary coolant	D ₂ O
Inlet temperature	25°C
Nominal outlet temperature	45°C
Upper outlet limit	55°C
Nominal mass flow	237 kg/s
Nominal volumetric flow	≈ 776 m ³ /h
Loop pressure regime	$\lesssim 3$ bar gauge
Core flow direction	top-to-bottom

5.3 Simplified secondary loop

The secondary side removes heat from the primary loop and rejects it to an ultimate heat sink. It is modelled at the minimum useful resolution using only a dynamic pipe with a fixed secondary temperature heat sink.

6 Core Thermal Model

6.1 Coolant channel

The primary loop represents the active region as one axially discretised Modelica dynamic pipe. It carries the full primary mass flow top-to-bottom and exposes one thermal connector per axial node. With ideal heat-port coupling, each heat port temperature is the local bulk coolant temperature used by the pipe energy balance.

The detailed distribution of flow among concentric gaps is not modelled. The current model therefore should not be interpreted as a hot-channel or margin-to-boiling calculation.

6.2 Fuel heat structure

Each axial coolant node is coupled to an aggregated RELAP-style one-dimensional fuel half-slab. The slab represents the path from the fuel meat symmetry plane to the coolant interface:

$$\left. \frac{\partial T_f}{\partial x} \right|_{x=0} = 0, \quad q_j'' = h_{core} (T_{wall,j} - T_{cool,j}).$$

The half-slab is discretised into $N_r = 5$ solid nodes. For axial node j and radial solid node r the Modelica model uses:

- `HeatCapacitor fuelNode[j,r]` for fuel heat storage,
- `PrescribedHeatFlow fuelHeat[j,r]` for deposited fission power,
- `ThermalConductor` links between adjacent radial fuel nodes,
- a half-cell wall conductor from the outer fuel node to the wall,
- `Convection coreConvection[j]` from the wall to the coolant heat port.

The heat-transfer coefficient is currently fixed,

$$h_{core} = 2.0 \times 10^4 \text{ W}/(\text{m}^2 \text{ K}),$$

so that the v1 heat path is tunable and independent of the effective pipe hydraulic area. Dittus-Boelter or a custom D₂O correlation can replace this fixed coefficient later, but that should be done with explicit hydraulic constants for the concentric coolant gaps.

6.3 Feedback temperatures

The fuel feedback temperature is a power-weighted axial average of the radial fuel-node average:

$$T_{f,eff} = \sum_{j=1}^{N_z} f_j \left(\frac{1}{N_r} \sum_{r=1}^{N_r} T_{f,j,r} \right),$$

where f_j are the normalised axial power fractions supplied to the Modelica model.

The moderator/coolant feedback temperature is the same axial weighting applied to the bulk coolant temperatures:

$$T_{m,eff} = \sum_{j=1}^{N_z} f_j T_{cool,j}.$$

In the current Modelica implementation $T_{cool,j}$ is taken from `core.heatPorts[j].T`. If a future toolchain exposes alias issues with those heat ports, the fallback approximation is

$$T_{m,eff} \approx \frac{T_{in} + T_{out}}{2}.$$

The moderator density feedback uses a compact D₂O density approximation rather than the current light-water medium density:

$$\rho_{D2O}(p, T) = \rho_{ref} [1 - \beta(T - T_{ref}) + \kappa(p - p_{ref})],$$

with $\rho_{ref} = 1105 \text{ kg/m}^3$, $T_{ref} = 300 \text{ K}$, $p_{ref} = 4.0 \times 10^5 \text{ Pa}$, $\beta = 3.0 \times 10^{-4} \text{ K}^{-1}$, and $\kappa = 4.5 \times 10^{-10} \text{ Pa}^{-1}$. The effective SI density is power-weighted over the same axial nodes:

$$\rho_{m,eff}^{SI} = \sum_{j=1}^{N_z} f_j \rho_{D2O}(p_j, T_j), \quad \rho_{m,eff} = \frac{\rho_{m,eff}^{SI}}{1000}.$$

Here $\rho_{m,eff}^{SI}$ is in kg/m^3 and $\rho_{m,eff}$ is in g/cm^3 for neutronics feedback interfaces that expect cgs density.

The model also exposes diagnostic maximum temperatures:

- `T_fuelCenterlineMax`, the maximum inner fuel-node temperature,
- `T_fuelWallMax`, the maximum fuel wall temperature.

7 Modelica Interface

The FMI-facing variables are:

Direction	Signal	Description
Python → Modelica	<code>totalPower</code>	total fission power [W]
Python → Modelica	<code>axialPowerFractions[n]</code>	normalised axial power fractions
Modelica → Python	<code>T_inlet</code>	core inlet bulk temperature [K]
Modelica → Python	<code>T_outlet</code>	core outlet bulk temperature [K]
Modelica → Python	<code>massFlow</code>	primary mass flow rate [kg/s]
Modelica → Python	<code>dp_core</code>	core pressure drop [Pa]
Modelica → Python	<code>T_fuelEff</code>	Doppler-relevant effective fuel temperature [K]
Modelica → Python	<code>T_moderatorEff</code>	effective moderator/coolant temperature [K]
Modelica → Python	<code>rho_m_eff_SI</code>	effective D ₂ O moderator density [kg/m ³]
Modelica → Python	<code>rho_m_eff</code>	effective D ₂ O moderator density [g/cm ³]
Modelica → Python	<code>T_fuelCenterlineMax</code>	diagnostic peak fuel centerline node [K]
Modelica → Python	<code>T_fuelWallMax</code>	diagnostic peak fuel wall [K]

The initial conditions are set near the nominal 20 MW operating point to avoid starting coupled point-kinetics runs from cold fuel. This is still an approximate initialization, not a formal steady-state solve.

The Python backend consumes `T_fuelEff`, `T_moderatorEff`, and `rho_m_eff` for point-kinetics reactivity feedback. The feedback slopes come from the OpenMC finite-difference reactivity-coefficient table under `openmc/reference_data/concentric/reactivity_coefficients`. At reset, the first available FMI effective state is used as the zero-feedback reference, so the coupled model starts from the same critical rod insertion as the rod-worth calibration. If the FMU is unavailable and the Python fallback thermal model is active, these feedback terms are not applied.

8 Deliberate Limits

The current implementation deliberately excludes:

- ring-by-ring or gap-by-gap coolant flow distribution,
- hot-channel factors and safety-margin calculations,

- temperature-dependent fuel properties,
- D₂O-specific medium replacement in Modelica,
- decay heat and shutdown cooling,
- active pool cooling and detailed reflector thermal hydraulics,
- pump coastdown and full pump head-curve dynamics.

These omissions keep the first coupled neutronics/thermal-hydraulics model small enough to verify while still providing the feedback temperatures needed for Doppler and moderator reactivity terms.